

Responsible Design Note

Scope Scout (Sawyer) · The Scope Scouts · v1 · 2026-05-28

READ-ONLY BY DESIGN

All Scope Scout actions against the Bloomreach workspace are read-only. The agent inspects workspace state through the Loomi Connect Marketing MCP and reasons over the results. It does not write data, modify configuration, create segments, configure journeys, or send messages. No write path exists in the runtime: every local tool is a get, list, search, or find operation, and every MCP tool maps to a read endpoint. This is the strongest single claim in the responsible-design posture and the property the rest of the design depends on.

RECOMMENDATION, NOT EXECUTION

Scope Scout produces a discovery agenda for architect review. It does not produce a scope, a quote, a campaign configuration, or a deliverable that can be sent to a client without human review. The architect reviews each finding, accepts or revises classifications, and decides what reaches the client.

| *The human owns the output.*

Architect review is a workflow property: the system surfaces evidence and structures decisions. The reasoning trace is fully visible so the architect can interrogate any classification before it becomes part of a client-facing artifact.

DATA HANDLING AND PERSISTENCE

In-session state — the conversation, the agent's intermediate reasoning, tool-call results — is held in memory at the running process level and discarded on session reset. Nothing is written to disk by Scope Scout in the normal reasoning loop. The architect may explicitly export a reasoning trace as a JSON artifact for review or record-keeping; this export is architect-initiated and visible, not a background persistence mechanism. No workspace data, customer records, or MCP response payloads are persisted by the agent beyond the active session.

CREDENTIALS AND ACCESS

No credentials are written to disk or logged at any level. The Anthropic API key is read from a process environment variable and never printed in error messages or logs (error messages name the variable, not the value). MCP authentication is handled by the mcp-remote bridge with the access token cached outside the application's working tree. No sk-ant... literal appears in any tracked source file, and the environment-variable file is excluded from version control. The codebase contains an explicit guard against logging the MCP whoami response, which embeds the live access token in its body.

ACCESS LOGGING

Every tool invocation — local knowledge or live MCP — emits a single structured access line to server-side stderr. Each line records the timestamp, tool name, source (local or live-mcp), query parameters, response size in bytes, status (ok or error), and latency. No response payloads are logged at the access level.

The access log is always on; it is the audit trail the responsible-design posture depends on. Verbose response logging (full response bodies) is gated behind a development flag (SCOPE_SCOUT_VERBOSE_LOG=1) and is disabled in all demo and production configurations. Full stack trace logging is not yet implemented; exceptions surface as status=error with message only, identified as a development-experience improvement for post-hackathon builds. The access log is not persisted to disk by Scope Scout itself; capture and retention are deferred to the host process.

SAFETY BOUNDS IN THE REASONING LOOP

The agentic reasoning loop is bounded at 25 tool iterations per session. If the agent has not produced a complete discovery agenda within 25 iterations, the loop terminates and surfaces what it has gathered for the architect to review rather than continuing indefinitely. The MCP layer detects rate-limit responses from the Loomi Connect endpoint (1 request per second) and backs off rather than retrying aggressively; live MCP calls are serialized with 1.2-second pacing to stay under the limit by design rather than relying on retry.

WHAT IS SIMULATED AND WHAT IS EXECUTED

The demo artifact at datadrovers.com/sawyer serves fixture data and makes no live MCP calls. It is a static surface that demonstrates the agent's output shape, reasoning trace, and discovery-agenda format using pre-captured workspace state. The live runtime, which performs real MCP introspection against the Loomi Connect

endpoint, runs only in authenticated local environments. Both modes share the same reasoning logic; the only difference is the source of workspace evidence (fixture vs live).

Production deployment would require UI authentication, session-scoped MCP token management, and access controls before any live workspace data is exposed to a non-architect user. These are identified as pre-production requirements, not current gaps.

WHAT WE DO NOT CLAIM

Scope Scout does not claim system-enforced architect override. The “propose, don’t decide” behavior is implemented in the agent’s system prompt and reinforced in the workflow; it is not a database-level or API-level write block beyond the read-only design above. We claim architect review as a workflow property and read-only inspection as a system property.

We do not claim that the system would prevent a malicious operator with code access from rewiring the agent to act on the workspace. The responsible-design posture is consistent with how it would be deployed and reviewed, not with adversarial-actor scenarios that any agent running with workspace credentials would face.

LIMITATIONS IDENTIFIED FOR PRE-PRODUCTION

- **Audit log retention** — Deferred to the host process; production deployment should pipe access lines to a managed logging surface with retention and search.
- **Anomaly detection** — Partial: the iteration cap is enforced and rate-limit hits are handled, but no broader anomaly-detection infrastructure (per-architect rate budgets, unusual-pattern alerts, drift detection) is in place.
- **Stack trace logging** — Not yet implemented; exceptions surface as status=error with message only. Identified as a development-experience improvement, not a security gap.
- **Token redaction** — The whoami access-token leak is guarded by an explicit logging exclusion, not by an MCP-layer policy. A future MCP version returning sensitive content in additional response bodies would need a more general redaction layer.
- **Demo timing fidelity** — Fixture mode runs without rate limits; demo recordings reflect fixture timing, not production-fidelity latency.

